

THE USE OF AI IN R

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Content

- AI is now part of the workflow
- Where AI helps an R beginner
- Tools overview 2026
- Live demo
- Example 1: decode an error message
- Example 2: starter code and critique
- AI for learning
- Risk 1: wrong or invented answers
- Risk 2: data privacy
- Ethics and disclosure
- Verify-everything habit
- Good vs bad prompts
- Key takeaways
- Suggested readings

AI is now part of the workflow

- AI assistants are already used by most analysts and students
- The question is no longer WHETHER to use AI
- The question is HOW to use it well
- Used well: faster, more learning
- Used badly: wrong results, privacy breach, integrity problems

HOWEVER

- Do not use AI for everything in R (end to end) - analogous giving a calculator to primary school students

Where AI helps an R beginner

- Explain error messages in plain language
- Write starter code for a task
- Explain unfamiliar functions and packages
- Suggest plot types and ggplot2 code
- Translate SPSS logic to tidyverse
- Always a starting point, never the final answer

Tools overview 2026

- ***GitHub Copilot: inside RStudio (1 year free for students)***
- ChatGPT, Claude, Gemini: web chat
- Posit's assistant in RStudio (need to pay)
- ***Google Colab AI assistant***

***somewhat free**

Live demo

- Google Colab AI assistant
- GitHub Copilot: inside RStudio

Example 1: decode an error message

- Run code and get an error
- Copy the full error message
- Paste it into the AI with your code
- Ask: 'Explain this error simply and how to fix it'
- Then check the fix yourself

Example 2: starter code and critique

- Ask for ggplot starter code for a boxplot
- Ask AI to critique your dplyr pipe
- Ask: 'Is there a simpler way?'
- Compare the AI answer with the documentation

AI for learning

- Ask: 'Explain this line like I am new to R'
- Ask AI to quiz you on what you learned
- Good for statistics CONCEPTS (p-value, confidence interval)
- Check concepts against a textbook

Risk 1: wrong or invented answers

- AI can invent functions and packages that do not exist
- Always check ?function and the package documentation
- AI does not know your study design unless you tell it
- Wrong test = wrong conclusion, even if the code runs
- Confident tone does not mean correct

Risk 2: data privacy

- NEVER upload identifiable patient data to public AI tools
- No names, IC numbers, MRN, phone, addresses, exact dates
- Also avoid small groups that could identify someone
- Public tools may store or reuse what you type
- Follow ethics approval and institutional data rules
- If unsure: use fake data or ask your supervisor

Ethics and disclosure

- Journals now require disclosure of AI use
- AI is NOT an author
- You are fully responsible for everything you submit
- Check your target journal's AI policy before submission

Verify-everything habit

- Run the code
- Check the output
- Sanity-check the numbers (sample size, ranges, direction)
- Compare with a simple method or documentation
- AI drafts, YOU are the analyst

Good vs bad prompts

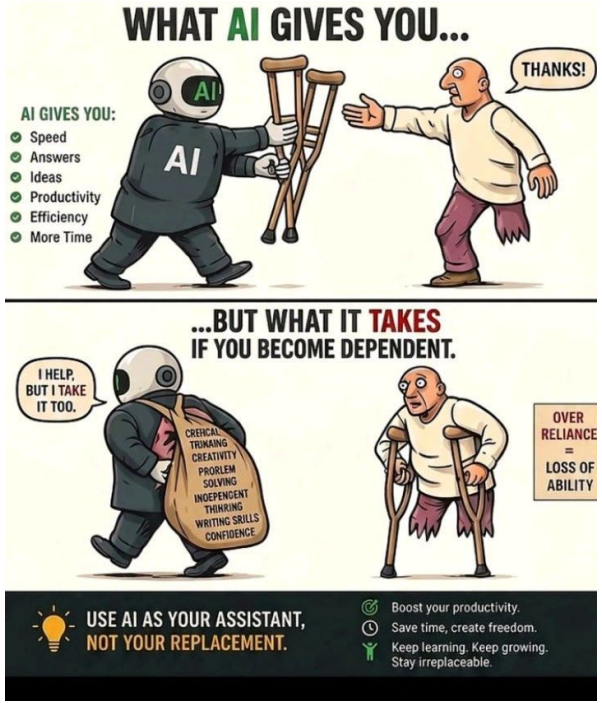
- BAD: 'Analyse my data'
- GOOD: 'My outcome is binary, exposure is smoking, 200 cases. I use R and the tidyverse. Which test and why?'

- BAD: 'Fix this' (no code, no error)
- GOOD: 'Here is my code and the full error. Explain it simply'

- BAD: pasting real patient rows
- GOOD: pasting a fake example with same columns

Key takeaways

- AI is a helper: you stay in charge
- Give context: data structure, goal, package
- Verify every function, test and number
- Protect patient privacy: no identifiable data
- Disclose AI use in theses and papers



Suggested readings

- Hanis, T. M. (2026). R essentials: https://tengku-hanis.github.io/r_essential/
- ICMJE: Recommendations on AI use in scholarly writing (icmje.org)
- Your target journal's instructions for authors (AI policy)



Any question?